

Stream Extraction Operator >>

`while( inFile >> number )` is rewritten as
`while( inFile.operator>>( number ) )` which returns
`while( inFile )` which is rewritten as
`while( inFile.operator bool() )` which is the same as
`while( !inFile.fail() )`

```
while( inFile >> number )
    cout << number << endl;
```

```
while( true )
{
    inFile >> number;
    if( inFile.fail() )
        break;
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inFile >> number;
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12

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12

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inFile.bad()	false

1	2	\n	3	4	\n	5	6	\n
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12\n
34\n
56\n

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inFile.bad()	false

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output


```
while( inFile >> str )
    cout << str << endl;
```

```
while( true )
{
    inFile >> str;
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
inFile >> str;
while( !inFile.eof() )
{
    cout << str << endl;
    inFile >> str;
}
```

```
while( !inFile.eof() )
{
    inFile >> str;
    cout << str << endl;
}
```

ab

ab

ab

ab

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

ab\n
cd\n
ef

File

Output

```
while( inFile >> str )
    cout << str << endl;
```

```
while( true )
{
    inFile >> str;
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
inFile >> str;
while( !inFile.eof() )
{
    cout << str << endl;
    inFile >> str;
}
```

```
while( !inFile.eof() )
{
    inFile >> str;
    cout << str << endl;
}
```

ab
cd

ab
cd

ab
cd

ab
cd

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

ab\n
cd\n
ef

File

Output

```
while( inFile >> str )
    cout << str << endl;
```

```
while( true )
{
    inFile >> str;
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
inFile >> str;
while( !inFile.eof() )
{
    cout << str << endl;
    inFile >> str;
}
```

```
while( !inFile.eof() )
{
    inFile >> str;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
ab
cd
```

```
ab
cd
```

```
ab
cd
ef
```

inFile.good()	false
inFile.eof()	true
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output

getline()


```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

ab

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

ab

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

ab

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

ab

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

ab\n
cd\n
ef\n

File

Output

```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
```

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

```
ab
cd
```

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

```
ab
cd
```

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
```

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

```
ab\n
cd\n
ef\n
```

File

Output

```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

```
ab
cd
ef
```

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

```
ab\n
cd\n
ef\n
```

File

Output

```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

```
ab
cd
ef
```

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

<code>inFile.good()</code>	false
<code>inFile.eof()</code>	true
<code>inFile.fail()</code>	true
<code>inFile.bad()</code>	false

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

```
ab\n
cd\n
ef\n
```

File

Output


```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

ab

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

ab

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

ab

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

ab

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

ab\n
cd\n
ef\n

File

Output

```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
```

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
ab
cd
```

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

```
ab
cd
```

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
```

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

```
ab\n
cd\n
ef\n
```

File

Output

```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

```
ab
cd
ef
```

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

```
ab\n
cd\n
ef\n
```

File

Output


```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

```
ab
cd
ef
```

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

inFile.good()	false
inFile.eof()	true
inFile.fail()	true
inFile.bad()	false

a	b	\n	c	d	\n	e	f	\n
---	---	----	---	---	----	---	---	----

```
ab\n
cd\n
ef\n
```

File

Output


```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

ab

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

ab

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

ab

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

ab

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

ab\n
cd\n
ef

File

Output

```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
```

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

```
ab
cd
```

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

```
ab
cd
```

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
```

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output

```
while( inFile.getline( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getline( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
inFile.getline( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getline( str, 4 );
}
```

```
ab
cd
ef
```

```
while( !inFile.fail() )
{
    inFile.getline( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

<code>inFile.good()</code>	<code>false</code>
<code>inFile.eof()</code>	<code>true</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output

```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.fail() )
        break;
    cout << str << endl;
}
```

```
ab
cd
ef
```

```
inFile.getLine( str, 4 );
while( !inFile.fail() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

```
ab
cd
ef
```

```
while( !inFile.fail() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

<code>inFile.good()</code>	<code>false</code>
<code>inFile.eof()</code>	<code>true</code>
<code>inFile.fail()</code>	<code>true</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output


```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

ab

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

ab

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

ab

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

ab

inFile.good()	true
inFile.eof()	false
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

ab\n
cd\n
ef

File

Output


```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
```

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
ab
cd
```

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

```
ab
cd
```

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
```

<code>inFile.good()</code>	<code>true</code>
<code>inFile.eof()</code>	<code>false</code>
<code>inFile.fail()</code>	<code>false</code>
<code>inFile.bad()</code>	<code>false</code>

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output

```
while( inFile.getLine( str, 4 ) )
    cout << str << endl;
```

```
ab
cd
ef
```

```
while( true )
{
    inFile.getLine( str, 4 );
    if( inFile.eof() )
        break;
    cout << str << endl;
}
```

```
ab
cd
```

```
inFile.getLine( str, 4 );
while( !inFile.eof() )
{
    cout << str << endl;
    inFile.getLine( str, 4 );
}
```

```
ab
cd
```

```
while( !inFile.eof() )
{
    inFile.getLine( str, 4 );
    cout << str << endl;
}
```

```
ab
cd
ef
```

inFile.good()	false
inFile.eof()	true
inFile.fail()	false
inFile.bad()	false

a	b	\n	c	d	\n	e	f	
---	---	----	---	---	----	---	---	--

```
ab\n
cd\n
ef
```

File

Output

If the state of streambuf is good, and the first character in streambuf is not '\n', then after the execution of

- `inFile.getLine`, '\n' is removed from streambuf if found,
- `inFile.getLine`, the state of streambuf is set to fail if '\n' is not found,
- `inFile >>`, '\n' is not removed from streambuf.

If the state of streambuf is good, and the first character in streambuf is '\n', then the execution of

- `inFile.getLine` removes '\n' from streambuf, reads an empty string and keeps the state good,
- `inFile >>` removes '\n' from streambuf, do correct read operation and keeps the state good.